



# INTRODUCTORY PAPER



# 2018 MATHEMATICS

**DO NOT OPEN THIS BOOKLET  
UNTIL INSTRUCTED.**

**30 QUESTIONS**

**TIME ALLOWED: 35 MINUTES**

STUDENT'S NAME: *Ng, L...*

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

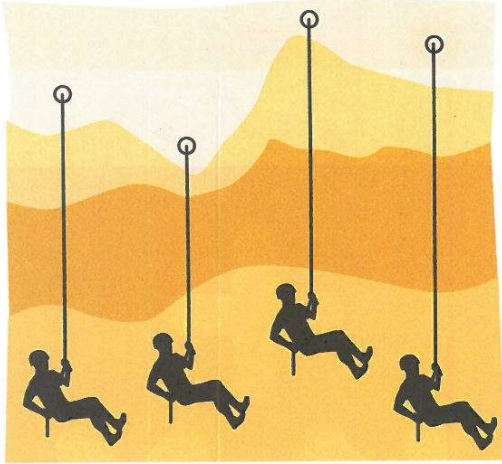
On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

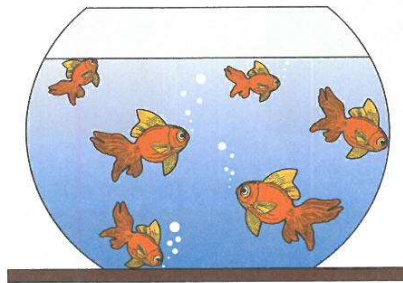
You are **NOT** allowed to use a calculator.

1. Which rope is the shortest?

(A) (B) (C) (D)



2. Toby had a glass bowl with these goldfish.

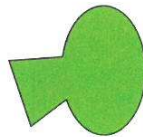


He put 4 more goldfish in the bowl.

How many goldfish did Toby have then?

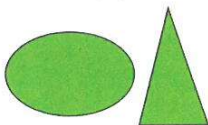
(A) 4 (B) 6 (C) 8 (D) 10

3. Emily put one shape on top of another shape. She drew around the edge to make this design.

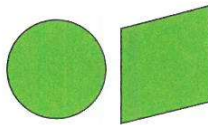


Which two shapes did Emily have?

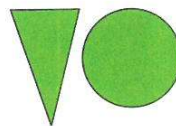
(A)



(B)



(C)



(D)





4. Lin's class counted the vehicles that went past the school in 5 minutes.  
They graphed their results.

**Vehicles Driven Past School**

|                                                                                   |                                                                                   |                                                                                   |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |                                                                                   |                                                                                   |                                                                                     |
|  |                                                                                   |  |                                                                                     |
|  |  |  |                                                                                     |
|  |  |  |  |
|  |  |  |  |

Which type of vehicle went past the school exactly 4 times in the 5 minutes?

- (A) (B) (C) (D)



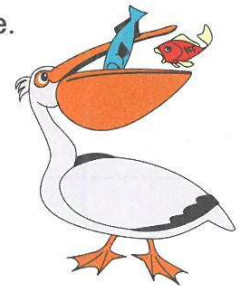
5. Ravi had 17 small fish in a bucket. He fed the pelican two fish at a time.

The number of fish left in the bucket after each feed followed this pattern:

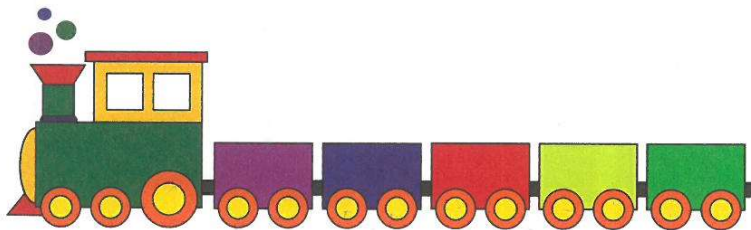
15, 13, 11, ?

What value must ? be?

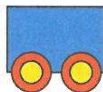
- (A) 8 (B) 9 (C) 10 (D) 12



6. Jack was drawing a train.



Then he drew another four carriages like this:



How many wheels did he draw in total?

- (A) 13 (B) 15 (C) 19 (D) 21

7. Lori goes to activities four afternoons a week. She has Art today.

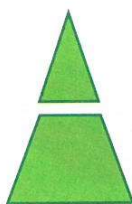
| Activity | Afternoon |
|----------|-----------|
| Art      | Tuesday   |
| Dance    | Friday    |
| Music    | Monday    |
| Netball  | Wednesday |

What activity will Lori do tomorrow afternoon?

- (A) Art                      (B) Dance                      (C) Music                      (D) Netball

8. Which of these was made by cutting a shape into equal parts?

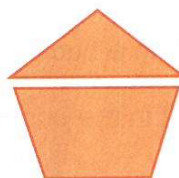
(A)



(B)



(C)

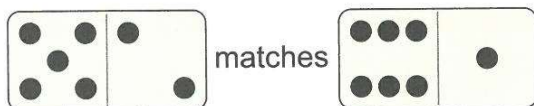


(D)

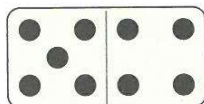


9. Jim is matching dominoes by adding their spots together.

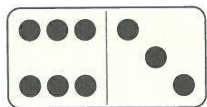
For example:



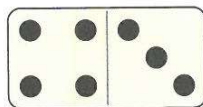
Jim picked up this domino.



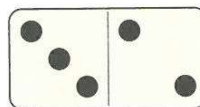
Which domino matches it?



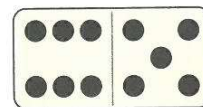
(A)



(B)



(C)



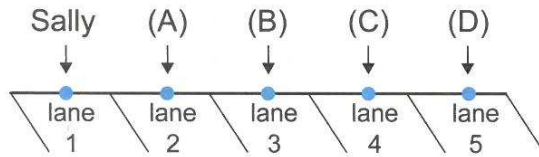
(D)

10. In a race, four students stand in their lanes in this order:

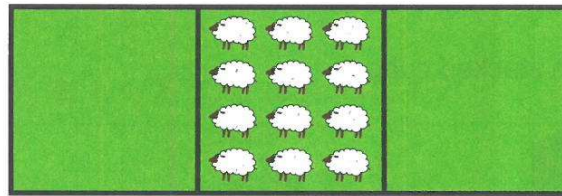
Sally, Jake, Tina, Amy.

There is one empty lane between Sally and Jake.

Where is Tina standing?



11. Farmer Jim has three sheep paddocks.

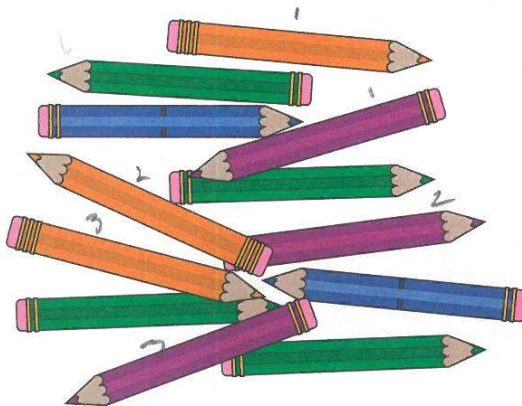


He wants each paddock to have the same number of sheep in it.

How many sheep does he need to move in total?

- (A) 3                      (B) 4                      (C) 6                      (D) 8

12. Chinli picked up one of these pencils without looking.



Which colour pencil was she most likely to pick up?

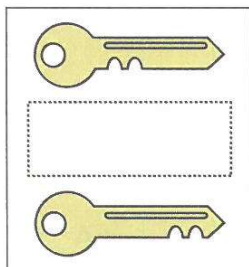
- (A)                       (B)   
(C)                       (D) 

13. Jai is using the number of cuts in the bottom edge of some keys to arrange them in order.

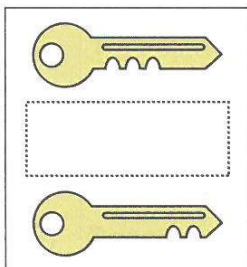


Between which two keys does this key belong?

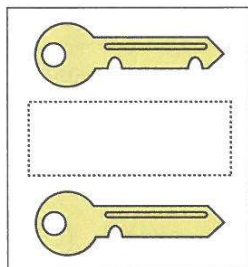
(A)



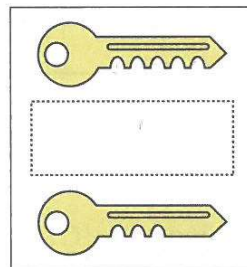
(B)



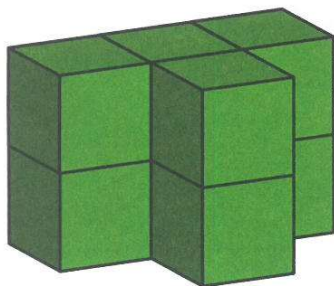
(C)



(D)



14. This solid is made from stacked cubes.



How many cubes make up this solid?

(A) 14

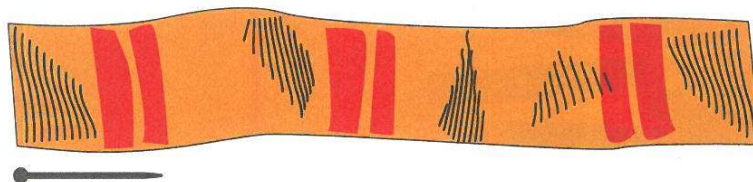
(B) 8

(C) 7

(D) 6

15. This picture shows a scarf that Zoe knitted.

She used a knitting needle to measure its length.



About how many knitting needles long is the scarf?

(A) 3

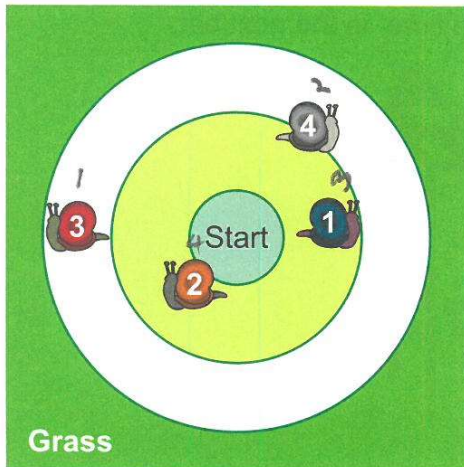
(B) 4

(C) 5

(D) 7



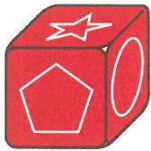
16. Four snails started in the centre circle. The picture shows their positions after 5 minutes.



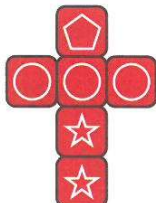
Which list gives the order of the snails, from the closest to the grass to the furthest from the grass?

- (A) 3, 4, 1, 2      (B) 3, 1, 4, 2      (C) 1, 3, 4, 2      (D) 1, 4, 3, 2

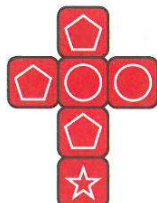
17. Jenny wants to make a dice.



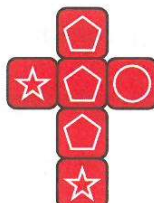
Which shape will fold to make a dice that gives Jenny the best chance of rolling a star?



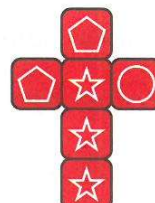
(A)



(B)



(C)



(D)

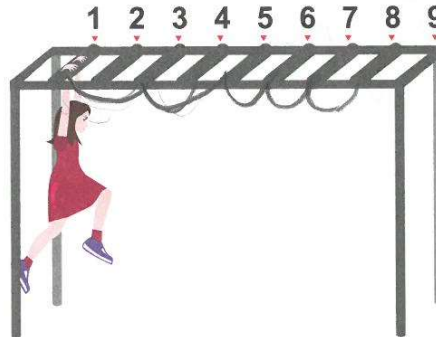
18. Tina lined up 6 toy cars. Each car was 5 centimetres long. She left a 2 centimetre gap between each car and the next.



How long was this line of cars?

- (A) 30 centimetres      (B) 32 centimetres  
(C) 40 centimetres      (D) 42 centimetres

19. Lily is swinging on the bars as shown. She starts with both hands on bar 1. It takes her 2 seconds to move both hands, one at a time, to the next bar.



From which bar is she hanging after 8 seconds?

- (A) 4 (B) 5 (C) 7 (D) 9

20. Mr Wong has two of these rewards cards. Each time he buys a loaf of bread he earns a stamp. When Mr Wong earns 10 stamps, he gets a free loaf of bread.



Mr Wong needs 3 more stamps to get his 4th free loaf of bread.

How many loaves of bread has Mr Wong bought?

- (A) 27 (B) 33 (C) 37 (D) 40

21. Each creature that Mike captures in his computer game scores a different number of points.

| Creature         |  |  |  |
|------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Number of points | 1                                                                                   | 5                                                                                    | 10                                                                                    |

Here are the creatures Mike captured.

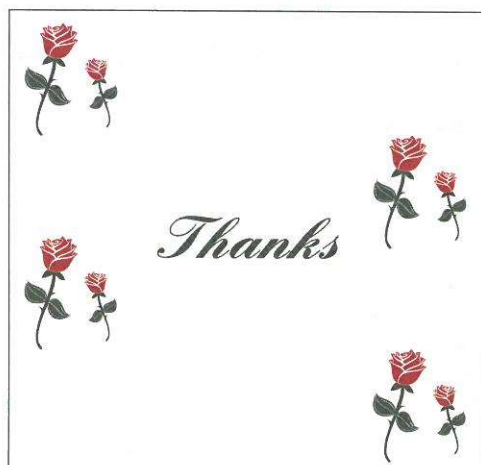


How many points did he score in total?

- (A) 6 (B) 16 (C) 23 (D) 27



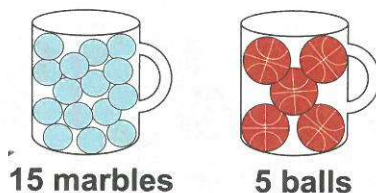
22. Rita made a card using a program on her computer. She began by drawing an image of two roses in the top left corner.



What operation did Rita use to move copies of the first image?

- (A) flip                      (B) slide                      (C) half turn                      (D) quarter turn

23. A mug can hold up to 15 marbles. The same mug can hold up to 5 balls.



A cake tin can hold up to 60 of these marbles.

About how many of these balls can the same cake tin hold?

- (A) 20                      (B) 40                      (C) 50                      (D) 80

24. Pete made up this number pattern.

|    |    |    |    |    |     |
|----|----|----|----|----|-----|
| 37 | 34 | 31 | 28 | 25 | ... |
|----|----|----|----|----|-----|

What are the next three numbers in the pattern?

- (A) 

|    |    |    |
|----|----|----|
| 21 | 18 | 15 |
|----|----|----|
- (B) 

|    |    |    |
|----|----|----|
| 22 | 17 | 14 |
|----|----|----|
- (C) 

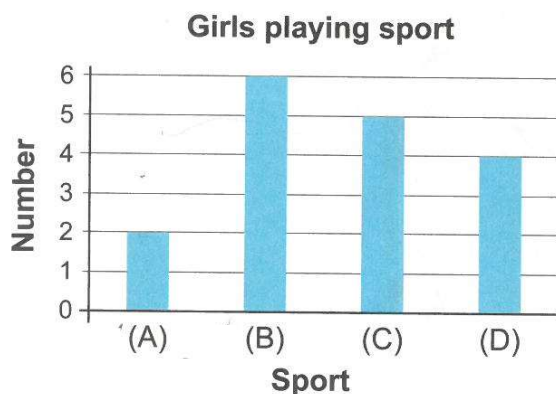
|    |    |    |
|----|----|----|
| 22 | 19 | 16 |
|----|----|----|
- (D) 

|    |    |    |
|----|----|----|
| 24 | 23 | 22 |
|----|----|----|

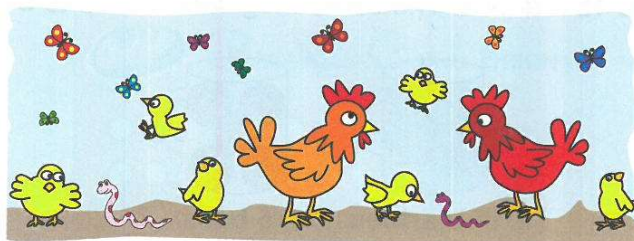
25. Each girl in Sasha's class plays only one sport.

Three more girls play tennis than play soccer.

Which column in this graph shows the girls who play tennis?



26. James is creating a table. He uses a different coloured dot to represent each chick, hen, worm and butterfly in this picture.



Which row is missing a dot?

|     | Animal | Number          |
|-----|--------|-----------------|
| (A) |        | ● ● ● ● ●       |
| (B) |        | ● ●             |
| (C) |        | ● ●             |
| (D) |        | ● ● ● ● ● ● ● ● |

27. Here is a pattern of shapes.



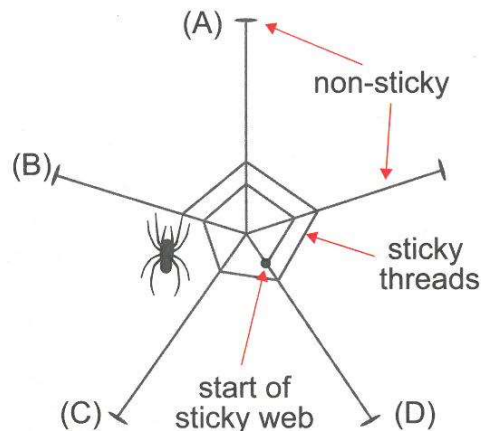
Which statement about this pattern is true?

- (A) There are fewer circles than triangles.
- (B) The circles are the largest shapes.
- (C) The shapes get smaller then get larger.
- (D) The squares are between the triangles.

28. Charlotte the spider is spinning a web. She joins 5 non-sticky threads at the centre. Then she spins 8 sticky threads between them as shown.

Charlotte stops spinning when there are 127 sticky threads in total.

On which non-sticky thread does Charlotte stop?



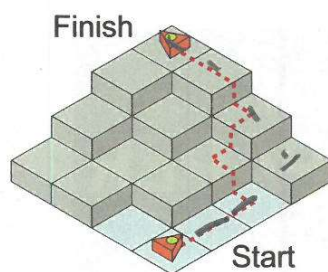
29. Arun and Sam are playing a board game. Together they have 28 tokens.

- Arun and Sam have the same number of tokens.
- There are 12 black tokens and the rest are red.
- Sam has 4 more red tokens than Arun.

How many black tokens does Arun have?

- (A) 4                      (B) 6                      (C) 8                      (D) 10

30. Sarah is playing a computer game. The picture shows how she moved her triangle from Start to Finish on the screen.



| Instruction  | Key to press |
|--------------|--------------|
| Move Forward | F            |
| Turn Left    | L            |
| Turn Right   | R            |
| Jump Up      | J            |

The keys she pressed were: F F L J R J L J F

Which keys will move the triangle from Start to Finish along a different path?

- (A) L F R J L F R J J L F                      (B) F F J L J J L F  
(C) L F J R F J J F                      (D) F L J F F J L J F



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### PRACTICE QUESTION

What is the missing number?

$$2 + 3 = ?$$

- (A) 2
- (B) 3
- (C) 4
- (D) 5

Turn your answer sheet over and fill in the correct oval.

### THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

|                                  |            |
|----------------------------------|------------|
| Australia                        | Year 2     |
| Brunei                           | Primary 2  |
| Egypt                            | Year 2     |
| Hong Kong                        | Primary 2  |
| Indian Subcontinent <sup>1</sup> | Class 2    |
| Indonesia                        | Year 3     |
| Malaysia                         | Standard 2 |
| Middle East <sup>2</sup>         | Class 2    |
| New Zealand/Pacific <sup>3</sup> | Year 3     |
| Singapore                        | Primary 1  |
| Southern Africa <sup>4</sup>     | Grade 2    |

<sup>1</sup> Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh.

<sup>2</sup> Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria, Jordan and Pakistan.

<sup>3</sup> Pacific Region: Vanuatu, Papua New Guinea and Fiji.

<sup>4</sup> Southern Africa Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia.

INTRODUCTORY  
**PAPER**



UNSW Global

## ANALYSIS BY QUESTION

The following table shows the questions you answered correctly ☒ and shades the questions that you answered incorrectly ☐. The questions are ranked from hardest at the top to easiest at the bottom.

|                     | Question content                                                                 | Area assessed       | Question number | Correct answer | Your answer | Students correct in AU |
|---------------------|----------------------------------------------------------------------------------|---------------------|-----------------|----------------|-------------|------------------------|
| Difficult Questions | Identify a set of directions to move from one position to another on a 3-D model | Space & Geometry    | 30              | A              |             | 18%                    |
|                     | Solve a two-step problem involving differences                                   | Number & Arithmetic | 29              | C              |             | 29%                    |
|                     | Find a missing element of a picture pattern that increases by ones               | Algebra & Patterns  | 13              | D              |             | 32%                    |
|                     | Count by tens to solve a problem                                                 | Number & Arithmetic | 20              | C              |             | 33%                    |
|                     | Solve a problem involving grouping in fours                                      | Number & Arithmetic | 11              | D              |             | 33%                    |
|                     | Count up by fives from a given starting point                                    | Number & Arithmetic | 28              | A              |             | 34%                    |
|                     | Solve a problem involving capacity                                               | Measures & Units    | 23              | A              |             | 34%                    |
|                     | Read a column graph to solve a problem                                           | Chance & Data       | 25              | C              |             | 36%                    |
|                     | Count by twos to determine the position after a given amount of time             | Number & Arithmetic | 19              | B              |             | 40%                    |
|                     | Position students according to a set of instructions                             | Space & Geometry    | 10              | C              |             | 41%                    |
| Easy Questions      | Match a picture to a table to solve a problem                                    | Chance & Data       | 26              | A              |             | 42%                    |
|                     | Read a table to identify the activity taking place tomorrow                      | Measures & Units    | 7               | D              |             | 43%                    |
|                     | Count by twos starting from three                                                | Number & Arithmetic | 6               | D              |             | 45%                    |
|                     | Measure the length of an object using informal uniform units                     | Measures & Units    | 15              | C              |             | 45%                    |
|                     | Identify the type of transformation used to move an image                        | Space & Geometry    | 22              | B              |             | 49%                    |
|                     | Use an array to solve a problem involving counting up by fives and twos          | Number & Arithmetic | 18              | C              |             | 54%                    |
|                     | Calculate the next three numbers in a decreasing number pattern                  | Algebra & Patterns  | 24              | C              |             | 55%                    |
|                     | Interpret a table to find a total score involving groups of ones, fives and tens | Algebra & Patterns  | 21              | D              |             | 62%                    |
|                     | Describe a pattern of shapes                                                     | Algebra & Patterns  | 27              | D              |             | 62%                    |
|                     | Identify the two shapes that have been overlapped                                | Space & Geometry    | 3               | A              |             | 63%                    |
|                     | Identify the most likely outcome                                                 | Chance & Data       | 12              | C              |             | 67%                    |
|                     | Solve a simple addition problem                                                  | Number & Arithmetic | 2               | D              |             | 72%                    |
|                     | Solve a problem involving a 3D object                                            | Space & Geometry    | 14              | B              |             | 75%                    |
|                     | Compare the number of dots on domino tiles                                       | Number & Arithmetic | 9               | A              |             | 78%                    |
|                     | Order distances from the circumference of a circle                               | Measures & Units    | 16              | A              |             | 79%                    |
|                     | Select the shape that gives the best chance of a particular outcome              | Chance & Data       | 17              | D              |             | 80%                    |
|                     | Complete a pattern decreasing by two each time                                   | Algebra & Patterns  | 5               | B              |             | 81%                    |
|                     | Interpret data in a picture graph                                                | Chance & Data       | 4               | C              |             | 81%                    |
|                     | Identify the shortest rope                                                       | Measures & Units    | 1               | B              |             | 85%                    |
|                     | Recognise a shape split into two equal parts                                     | Number & Arithmetic | 8               | B              |             | 91%                    |